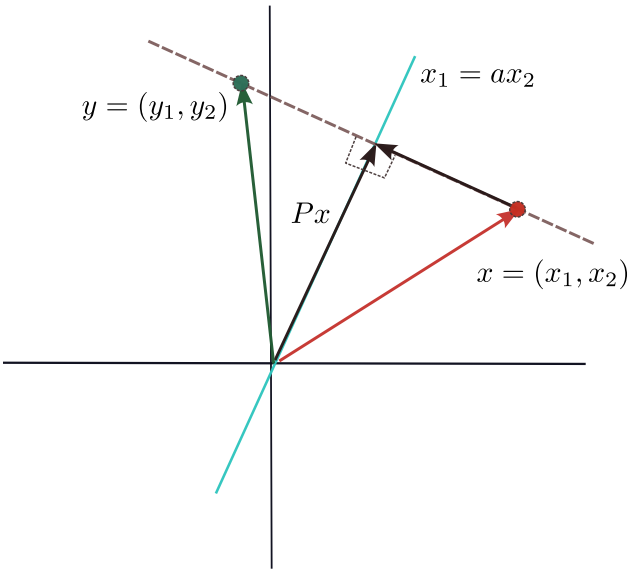
Reflection About a Line

Consider a line $x_1 = ax_2$ in $\mathbf{R}^2$ where a constant $a \in \mathbf{R}$ is given. Given a vector x and we aim to find the reflection of x about the line $x_1 = ax_2$. The result of the reflection vector is denoted as $y = T(x)$, which is viewed as a transformation on x. You can think of the principle of reflection as that the point Px is the projected vector of x on the line. Then you extend the point Px on a certain direction with the distance that is equal to the distance from x to Px.



Code a function that takes the original input vector x and the line parameter a. The function returns y, the reflection.

Input Format

- an integer a, $-10 \leq a \leq 10$
- a vector $x \in \mathbf{R}^2$

Output Format

- $y \in \mathbf{R}^2$: The reflection.

Implementation

Goal: Fill in the following function:

```
import numpy as np

def reflection_aboutline(x, a):
    ...
    return y, Px

if __name__ == "__main__":
    import matplotlib.pyplot as plt
    ...  # try to plot the result here
```